COURSE NLP301 / MODULE NL

Text Representation

Learning goal: explain core concepts, compare related methods, and apply them to realistic machine learning or educational technology scenarios.

Document ID: NLP301-02_Text_Representation | Format: instructional Q&A | Revision: 1.0

01. How does bag of words represent a document?

Bag of words counts vocabulary terms in a document, ignoring most order. It is simple and interpretable but treats related terms as unrelated and often produces sparse vectors. In practice, it helps to think of this as a sequence of steps rather than a single operation. Each step affects the information available to the next stage, so implementation choices should be validated on representative data. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

02. What is TF-IDF?

TF-IDF weights a term by its frequency in a document and reduces the weight of terms common across many documents. It supports strong lexical baselines and rare technical terms; vocabulary statistics must be fitted on the training corpus. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

03. What is one-hot encoding for words?

Each vocabulary item maps to a vector with one active position. Different words have orthogonal vectors regardless of meaning, and representation size grows with the vocabulary. Learned embeddings address some of these limits. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

04. How do Word2Vec CBOW and skip-gram differ?

CBOW predicts a center word from surrounding words; skip-gram predicts surrounding words from a center word. Both learn dense vectors from co-occurrence, but assign one representation per word regardless of sentence context. A useful way to understand how do word2vec cbow and skip-gram differ is to compare the methods not only by definition, but also by the kind of input they expect, the output they produce, and the situations in which each is useful. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

05. What is GloVe?

GloVe learns word vectors from aggregate global word co-occurrence statistics. It is a static embedding method: the same word receives the same vector in different sentences, which can blur multiple meanings. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

06. What are contextual embeddings?

A contextual model creates token representations conditioned on surrounding text, so 'bank' in finance can differ from 'bank' beside a river. These are useful for language understanding but cost more computation than sparse vectors. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.

07. What is a sentence embedding?

A sentence embedding compresses a whole sentence or passage into a dense vector intended to preserve semantic similarity. Retrieval quality depends on the embedding model, text granularity, domain, and distance measure. This concept is important because it connects the underlying theory with decisions made when building a real system. Understanding both its purpose and its limitations helps avoid treating the technique as a rule that works equally well in every dataset. For example, in an educational technology system, the same idea can be tested using course content, student questions, assessment records, or retrieved passages. A team should define the expected behavior first, measure it on held-out examples, and inspect failures rather than relying only on an aggregate score. For a production or RAG-oriented workflow, preserve useful metadata and reproducible preprocessing settings so that an unexpected result can be traced back to its source. This also makes it easier to compare alternative configurations and determine whether an apparent improvement comes from better modeling, better retrieval, or simply a change in the evaluation data. An important limitation is that no single technique or metric guarantees good real-world performance. Data leakage, distribution shift, poorly chosen evaluation examples, and mismatched assumptions can all make results look stronger than they really are. The practical approach is therefore to establish a baseline, change one major factor at a time, and verify the effect using representative validation data.